

2025 AMC 10A Problem 2 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10A Problem 2. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10A Problem 2

**Problem:**

A box contains 10 pounds of a nut mix that is 50 percent peanuts, 20 percent cashews, and 30 percent almonds. A second nut mix containing 20 percent peanuts, 40 percent cashews, and 40 percent almonds is added to the box resulting in a new nut mix that is 40 percent peanuts. How many pounds of cashews are now in the box?

Answer Choices:

- A. 3.5
- B. 4
- C. 4.5
- D. 5
- E. 6

Solution:

Let the weight of the second mix be x pounds.

We start by finding the equation for the number of peanuts. We get

$$\begin{aligned}\text{Peanuts from first mix} &= 10(0.50) = 5, \\ \text{Peanuts from second mix} &= x(0.20) = 0.2x.\end{aligned}$$

Therefore, in the final mixture of $10 + x$ pounds, we know that peanuts are 40% of the mixture, so we can setup the following equation to get:

$$\frac{5 + 0.2x}{10 + x} = 0.4$$

We can solve the equation to get $x = 5$.

Now, let's find the number of cashews in the final mixture. We know

$$\begin{aligned}\text{Cashews from first mix} &= 10(0.20) = 2, \\ \text{Cashews from second mix} &= 5(0.40) = 2, \\ \text{Total cashews} &= 2 + 2 = 4.\end{aligned}$$

so the final mixture contains 4 pounds of cashews, for an answer of **(B) 4**.

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